



**BITS Pilani**  
Pilani Campus



# **SURVEYING (CE F213)**

## **(Plane Table Surveying)**

# Plane Table Surveying

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Plane tabling(Plane Table Surveying) is a graphical method of surveying where the **field work** and **plotting** are done **simultaneously** in the field and such survey does not involve the use of a **field book(data entry book)**.

## To learn about

Principle of Plane Table surveying

Plane Table and related accessories

Adjustment of the Plane Table

Different methods of Plane Table Surveying

## Principle of Plane Table surveying

The principle of plane tabling is **parallelism**, meaning that the rays drawn from stations to objects on the paper are parallel to the lines from the stations to the objects on the ground.

The relative positions of the objects on the ground are represented by their plotted positions on the paper and lie on the respective rays.

The table is always placed at each of the successive stations parallel to the position it occupied at the starting station.

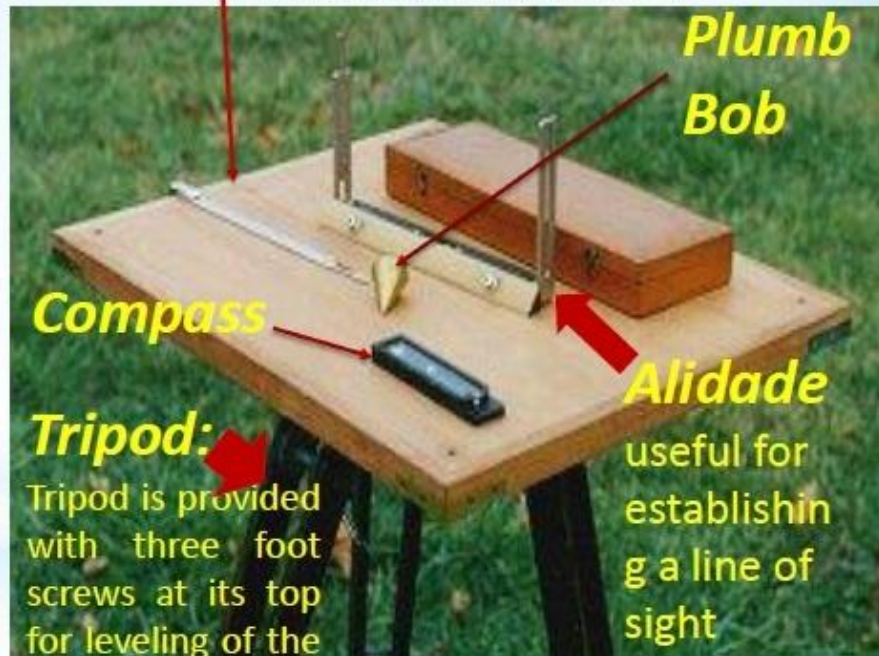
## Plane Table and related accessories

1. Plane table
2. Tripod
3. Alidade (Plain Alidade, Telescopic Alidade)
4. Spirit Level
5. Compass (Through compass, circular box compass)
6. Plumbing fork(U-fork) with plumb bob
7. Pegs, Wooden hammer
8. Tape
9. Ranging rods
10. Drawing Paper
11. Water cover

# Plane Table Surveying

## Plane Table and related accessories

**Plan Table:** (drawing board made from well-seasoned wood with its upper surface is exactly planed. It is normally rectangular in shape with size 75 cm x 60 cm)



## Plane table



- The plane table is a drawing board of size 750 mm x 600 mm made of well seasoned wood like teak, pine etc.
- The top surface of the table is well leveled.
- The bottom surface consists of a threaded circular plate for fixing the table on the tripod stand by a wing nut.
- The plane table is meant for fixing a drawing sheet over it.
- The positions of the objects are located on this sheet by drawing rays and plotting to any suitable scale.



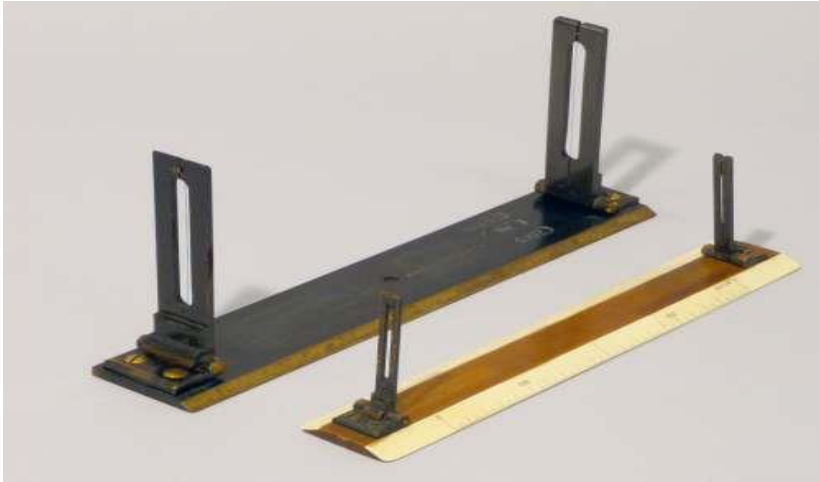


## Tripod (used for fixing the table)





Alidade (Plain Alidade, Telescopic Alidade) – used to establish a line of sight

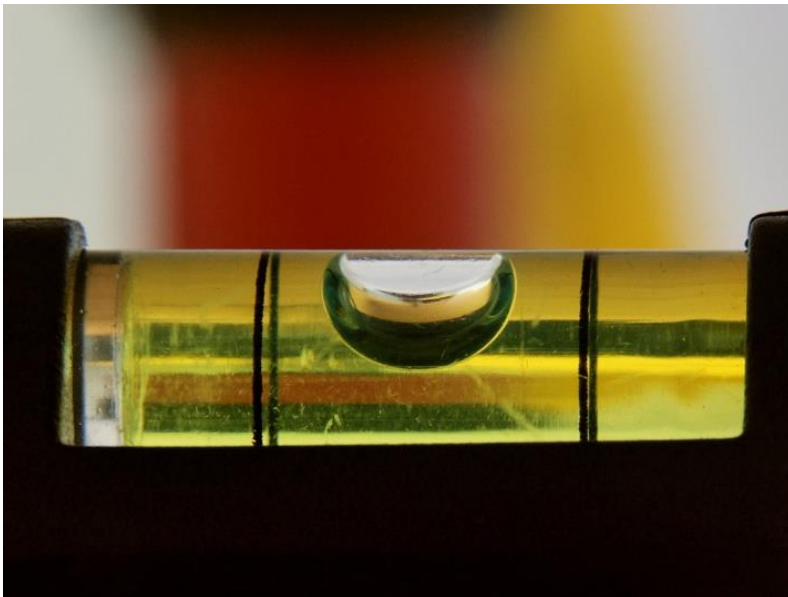


Plain Alidade



Telescopic Alidade

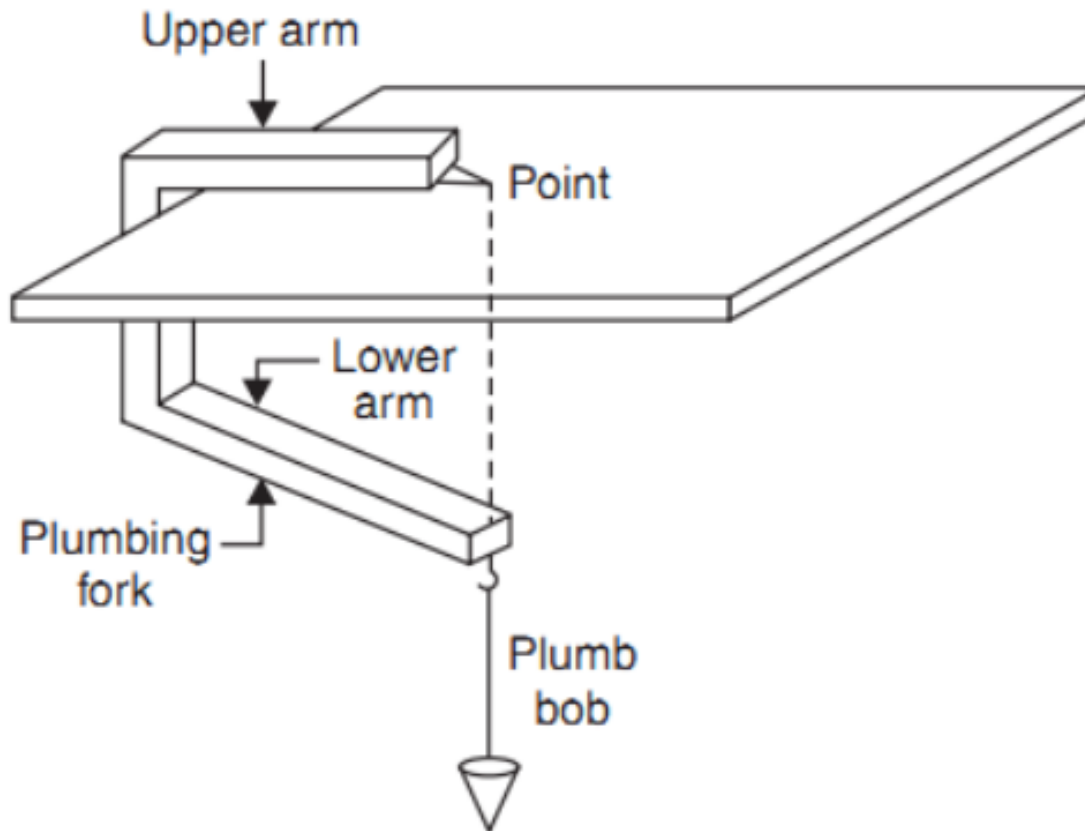
## Spirit Level used for levelling of Table



## Compass



## Plumbing fork(U-fork) with plumb bob



# Plane Table surveying

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## Adjustment of the Plane Table

- Fixing the plane table on the tripod stand
- Leveling the table
- Centering the table
- Marking the North-line
- Providing proper **Orientation** to the table

**Orientation**- The table should be parallel in all stations, i.e. Orientation of the table should be identical in all stations.

## Methods of Plane Table Surveying

1. Radiation Method
2. Intersection Method
3. Traversing Method
4. Resection Method

### Special Methods of Resection are:

- I. Compass Method
- II. Back sighting Method
- III. Two-point problem
- IV. Three-point problem

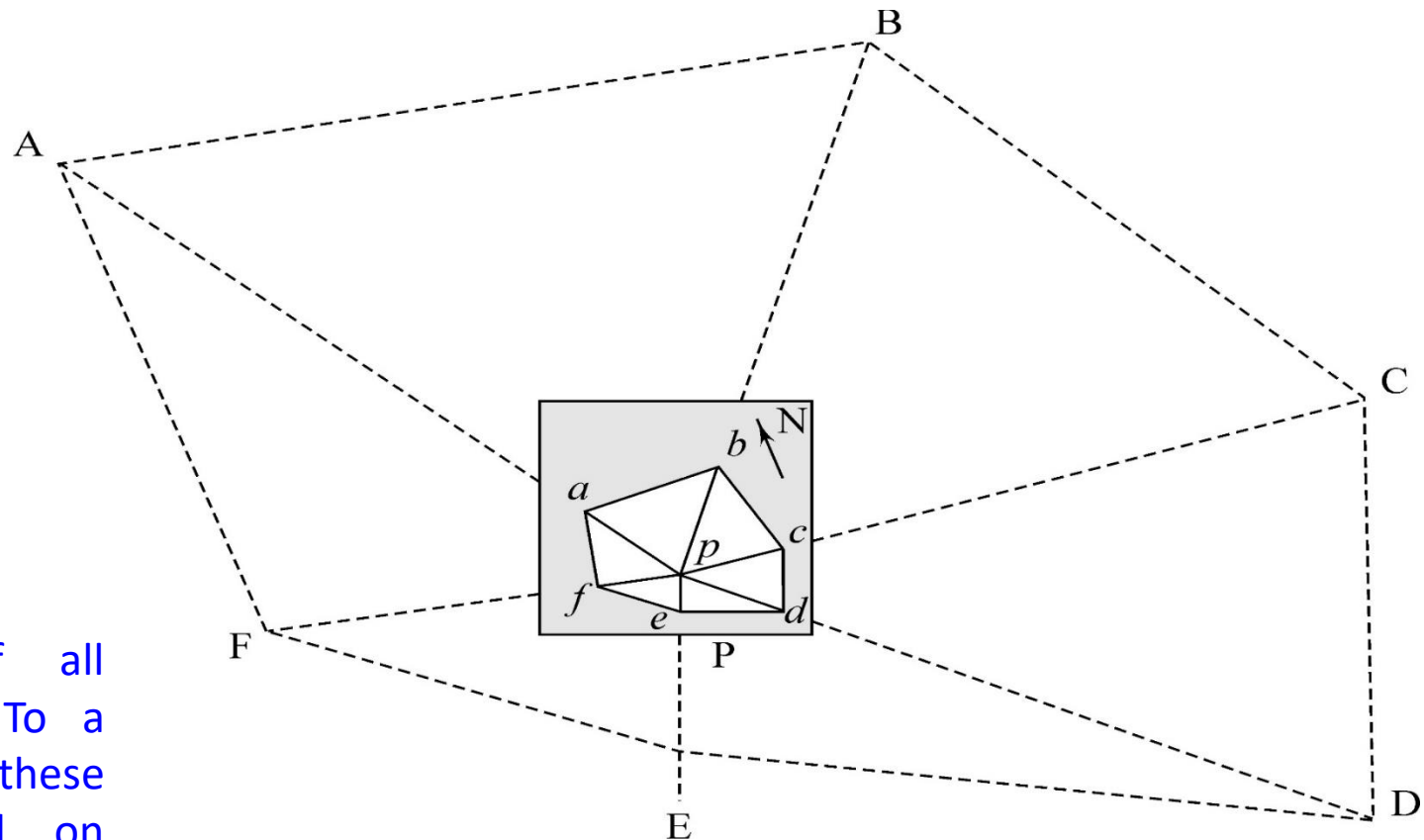
### Different methods of Three-point problem

- a) Tracing Paper method(Mechanical Method)
- b) Graphical method(Bessel's Method)
- c) Trial and error method(Lehmann's method)



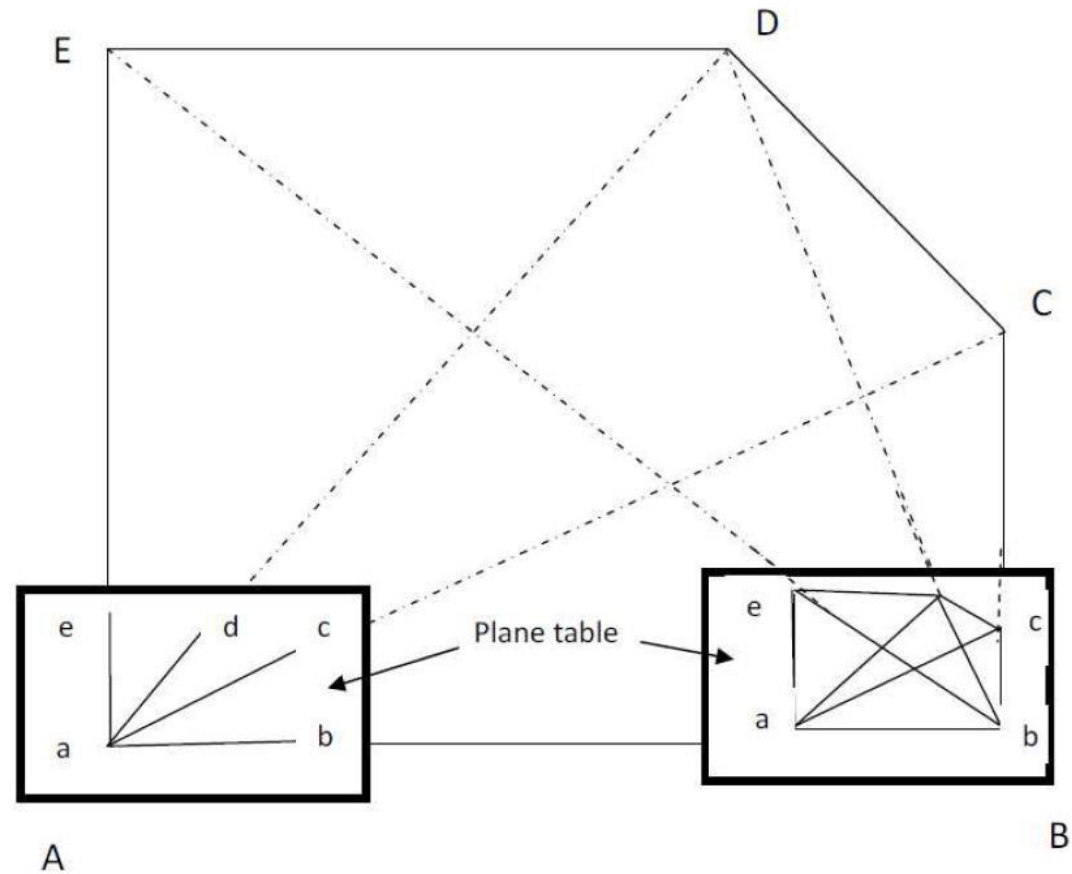
## Radiation Method

Measurement of all lines are done. To a suitable scale, these lines are marked on the drawing sheet.

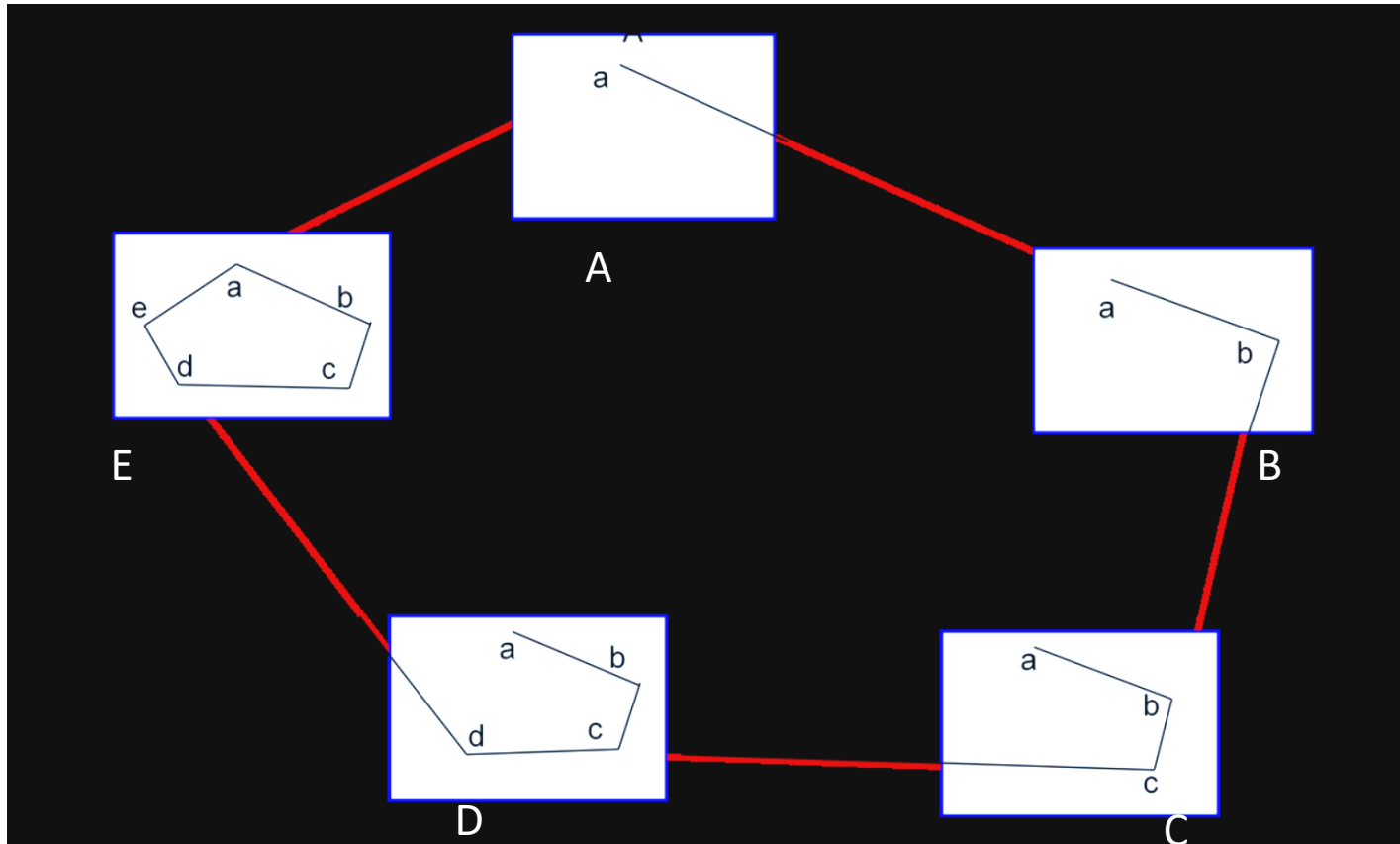


## Intersection Method

AB is the base line. The base line measurement is done. To a suitable scale AB is marked as ab on the drawing sheet. Other points are found from intersection



## Traversing Method



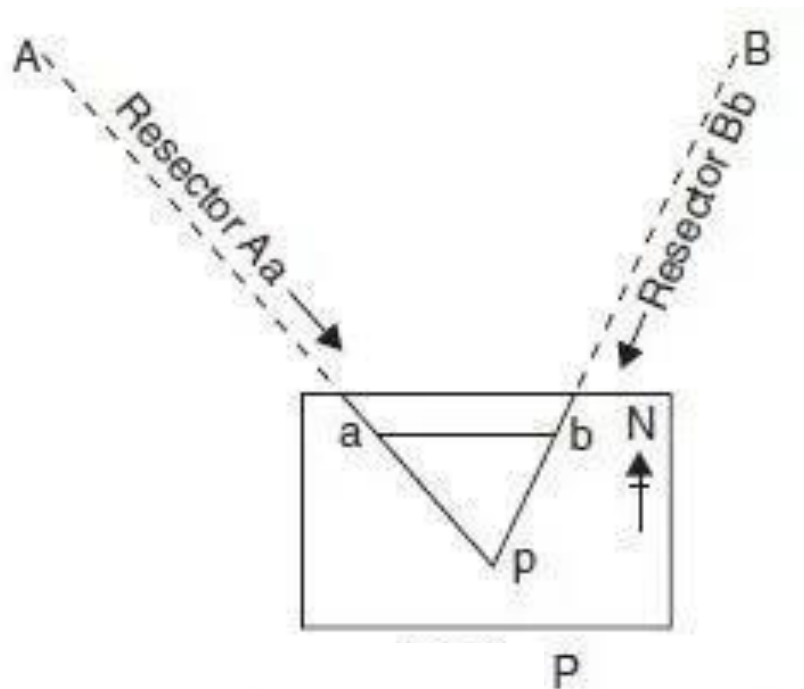
## Resection Method

It is a method of Orientation. It is used when the table takes a position on the ground is not located on the drawing sheet.

It is the process of locating the table position on the drawing sheet by drawing rays from the known stations whose positions are already marked on the sheet. The resection of the rays will give the position of the table on the drawing sheet if the orientation of the table is correct. ( i.e. if the orientation of the table becomes identical to the previous orientation while the known positions are marked on the sheet.)

The rays are called **resectors**.

Resection by Compass ( The North direction is taken as the reference in this case)



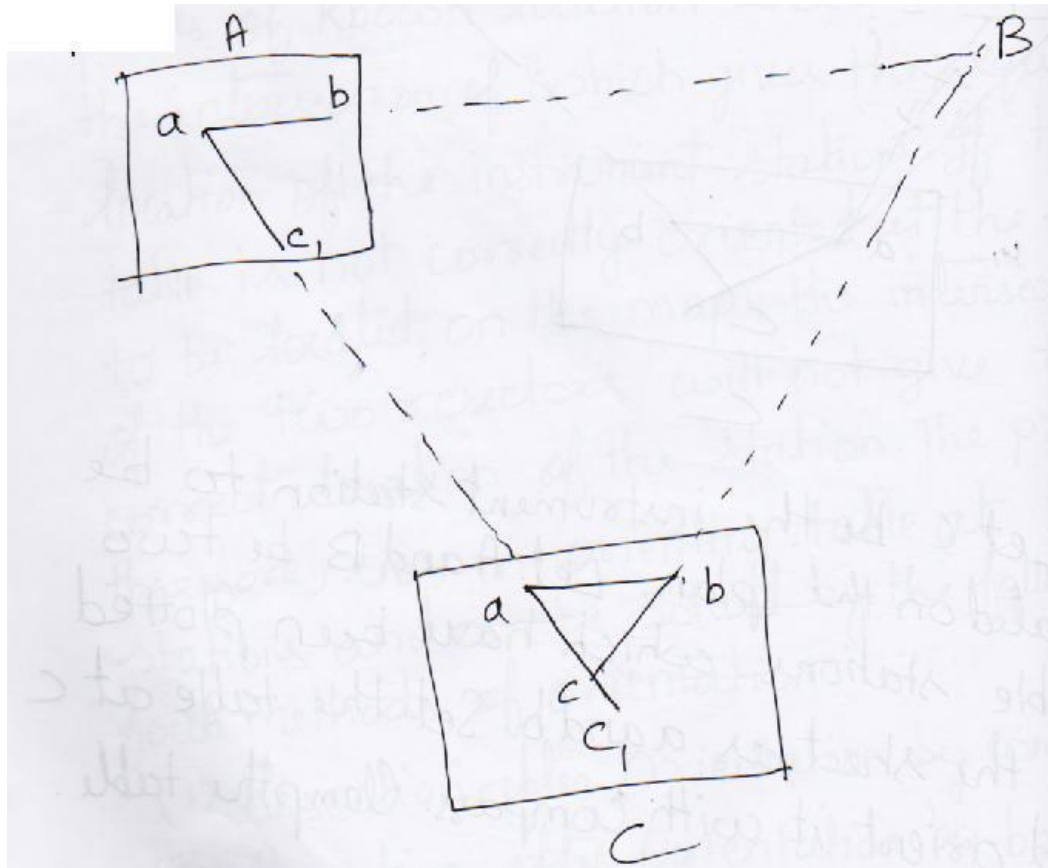
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Let  $a$  and  $b$  be the plotted positions of  $A$  and  $B$  of two well defined points in the field.

Keeping the through compass along north direction marked on the drawing sheet table is oriented on station  $P$ , the position of which is to be found on paper.

The resectors  $Aa$  and  $Bb$  are drawn to locate  $p$  the plotted position of station point  $P$ . This method gives satisfactory results, if the area is not influenced by local attractions.

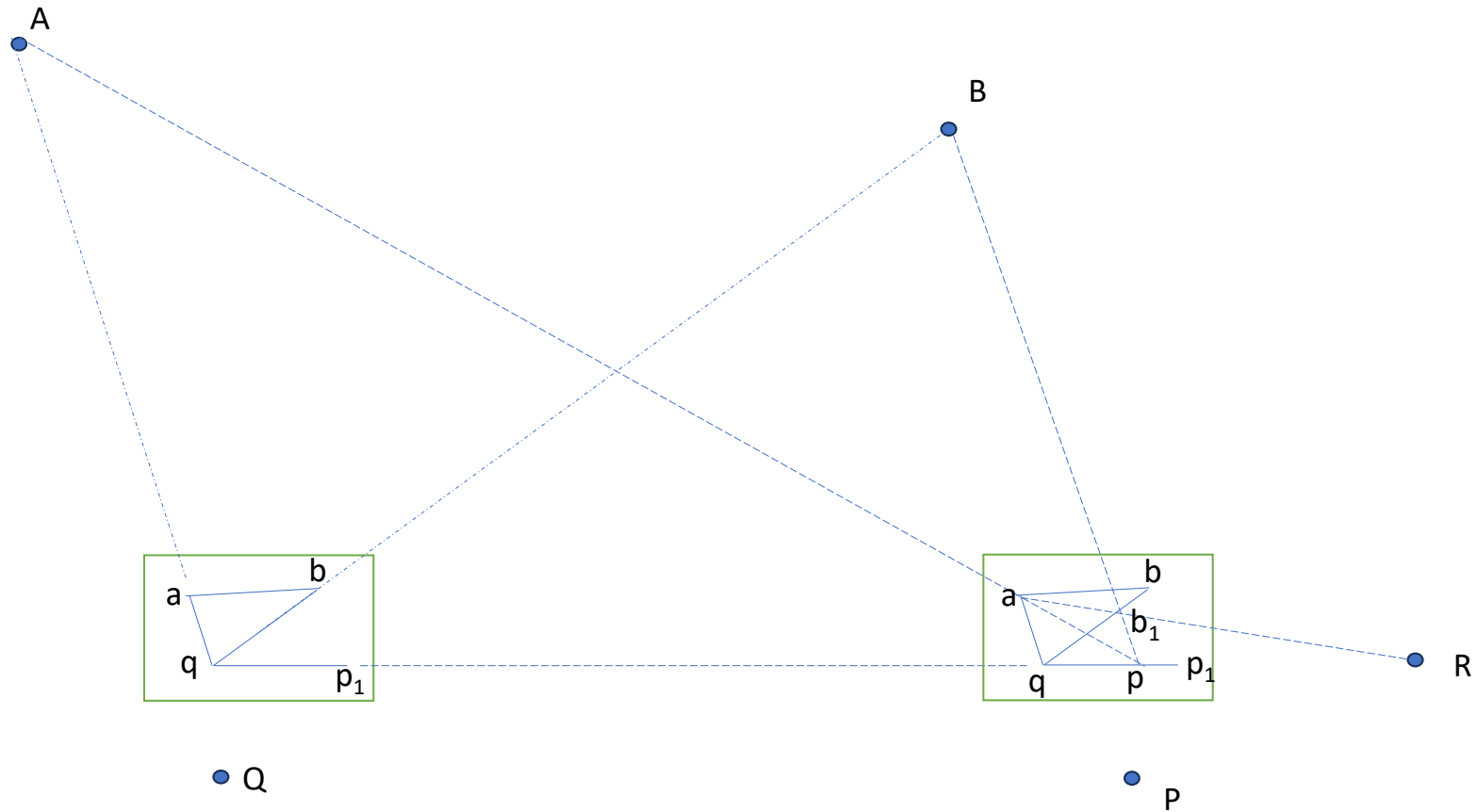
## Resection by Back sighting Method





- Let C be the station on the ground to be located on the drawing sheet. 'A' and 'B' be two visible ground points which are already plotted on the sheet as 'a' and 'b'.
- To set the table at 'A' and orient it by back sighting 'B' along 'ab'.
- Pivoting the alidade at 'a'. sight 'C' and to draw a ray. Estimate roughly the position of 'C' on this ray as  $c_1$ .
- Shift the table to 'C' and center it approximately with respect to  $c_1$ . Keep the alidade on the line  $c_1a$  and orient the table by back sight to 'A', Clamp the table which has been oriented.
- Pivoting the alidade about 'b', sight 'B' and draw the resector 'bB' to intersect the ray ' $c_1a$ ' in 'c'. Thus, 'c' is the location of the instrument station.

## Resection by Two point Method



Let A and B be two well defined points and their positions a and b are already known.

Now the problem is to orient the table at P(ground station) so that its position p(on the sheet) can be obtained.

The following steps may be followed to solve this problems:

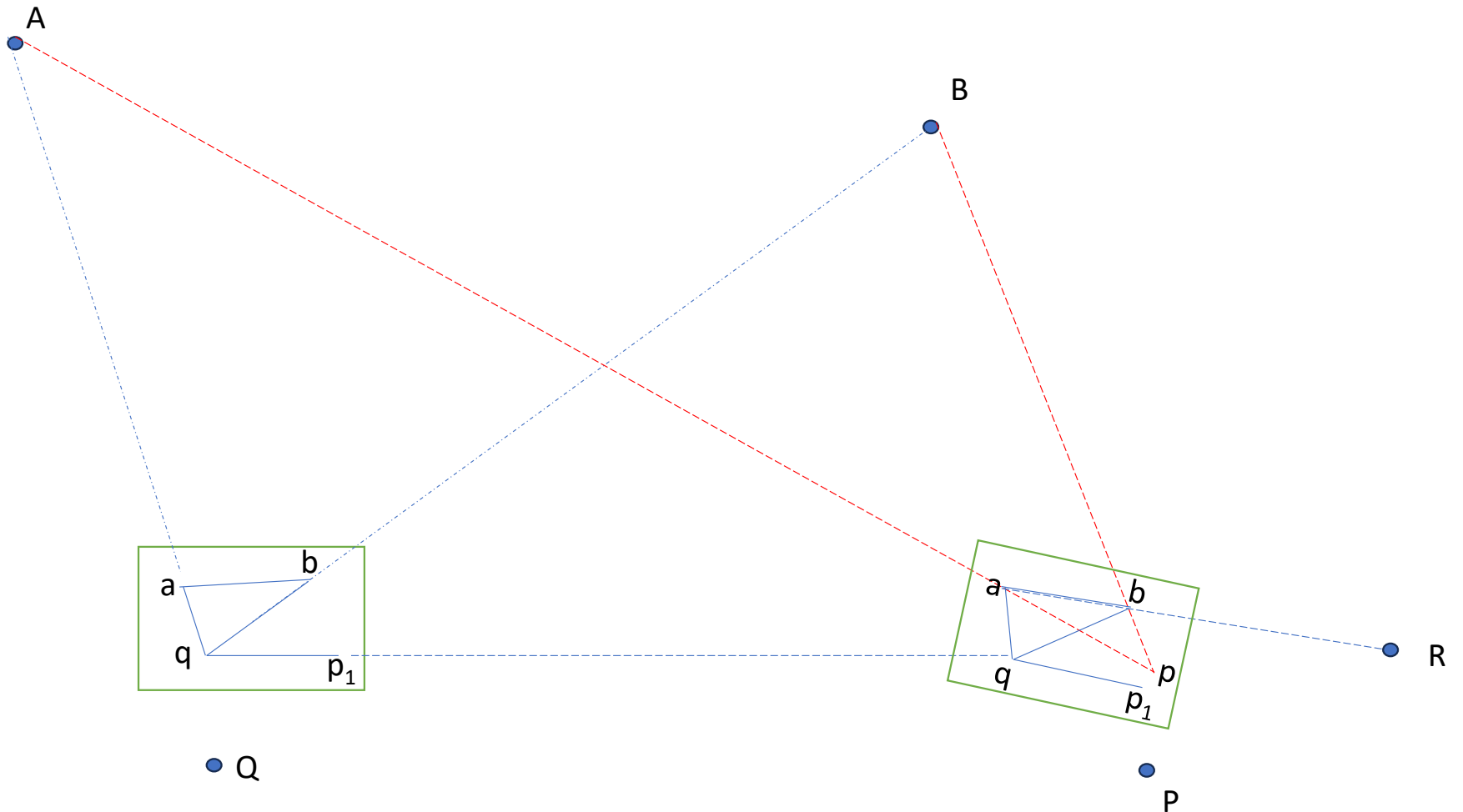
- (i) To select a suitable point Q near P such that QP and AB are approximately parallel and the angles PAQ and PBQ are not acute.
- (ii) To orient the table at Q such that AB and ab are approximately parallel and to draw the resectors Aa and Bb to get the point q. This q is not the correct position of Q as the orientation is not exact.
- (iii) To draw the ray qp and locate  $p_1$  with estimated distance QP.
- (iv) To shift the plane table to P and to orient the table by back sighting to Q.
- (v) To draw the resector Aa to get p.

(vi) To draw the ray  $pB$ . If the ray passes through  $b$  then the orientation is correct and  $p$  is the correct position of  $P$ . If not the ray( $pB$ ) intersect line  $bq$  at  $b_1$ .

(vii) The points  $b$  and  $b_1$  are not coinciding due to the angular error in the orientation of table. The angle  $bab_1$ , is the angular error in orientation. To correct it,

Lets fix a ranging rod at  $R$  along  $ab_1$ , Lets unclamp the table and rotate it till line  $ab$  sights ranging rod at  $R$ . Then clamp the table. This gives the correct orientation of the table which was used in plotting the points  $A$  and  $B$ .

(viii) The resectors  $Aa$  and  $Bb$  are drawn to get the correct plotted position  $p$  of the station  $P$ .

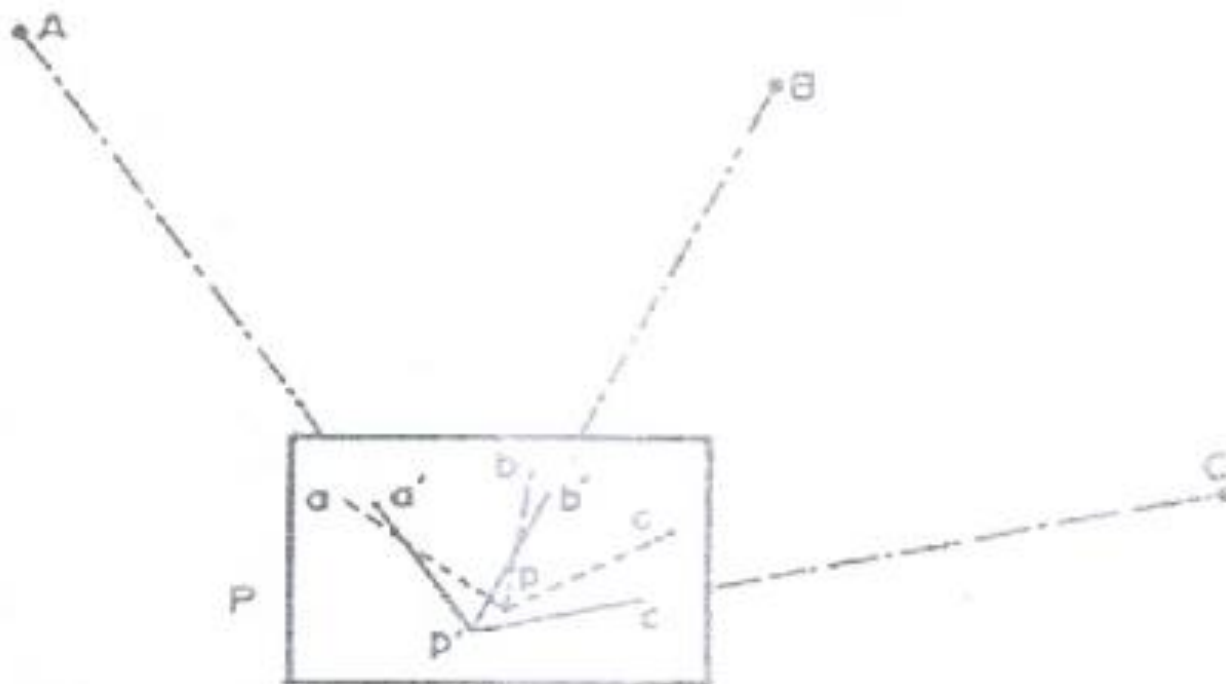


## Resection by Three point Method

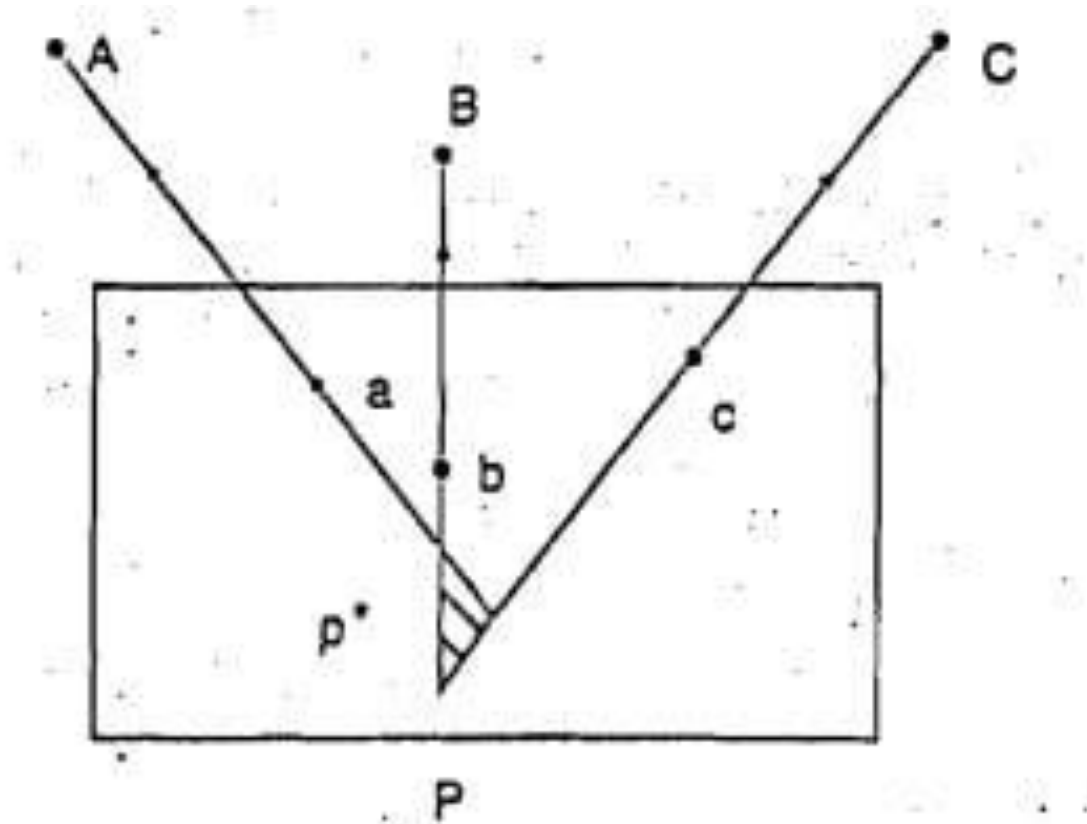
The process to locate the ground station on to the drawing sheet using observations to three well defined points whose positions are already known on the sheet. The different methods of three point problem are :

- a) Tracing Paper method(Mechanical Method)
- b) Graphical method(Bessel's Method)
- c) Trial and error method(Lehmann's method)

## Tracing Paper method(Mechanical Method)



## Trial and error method(Lehmann's method)

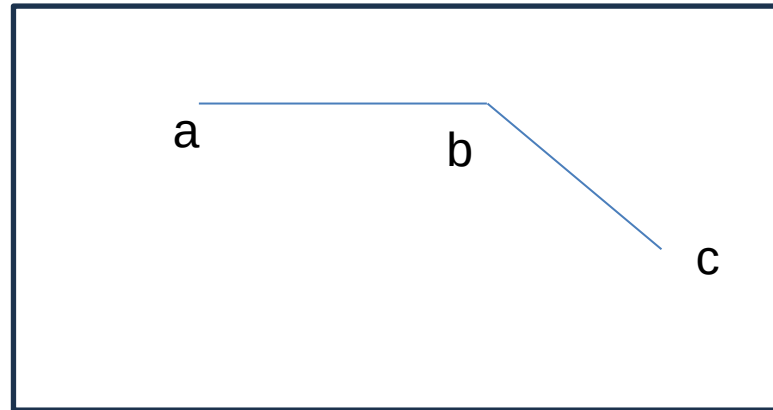




## Graphical Method(Bessel's Method)

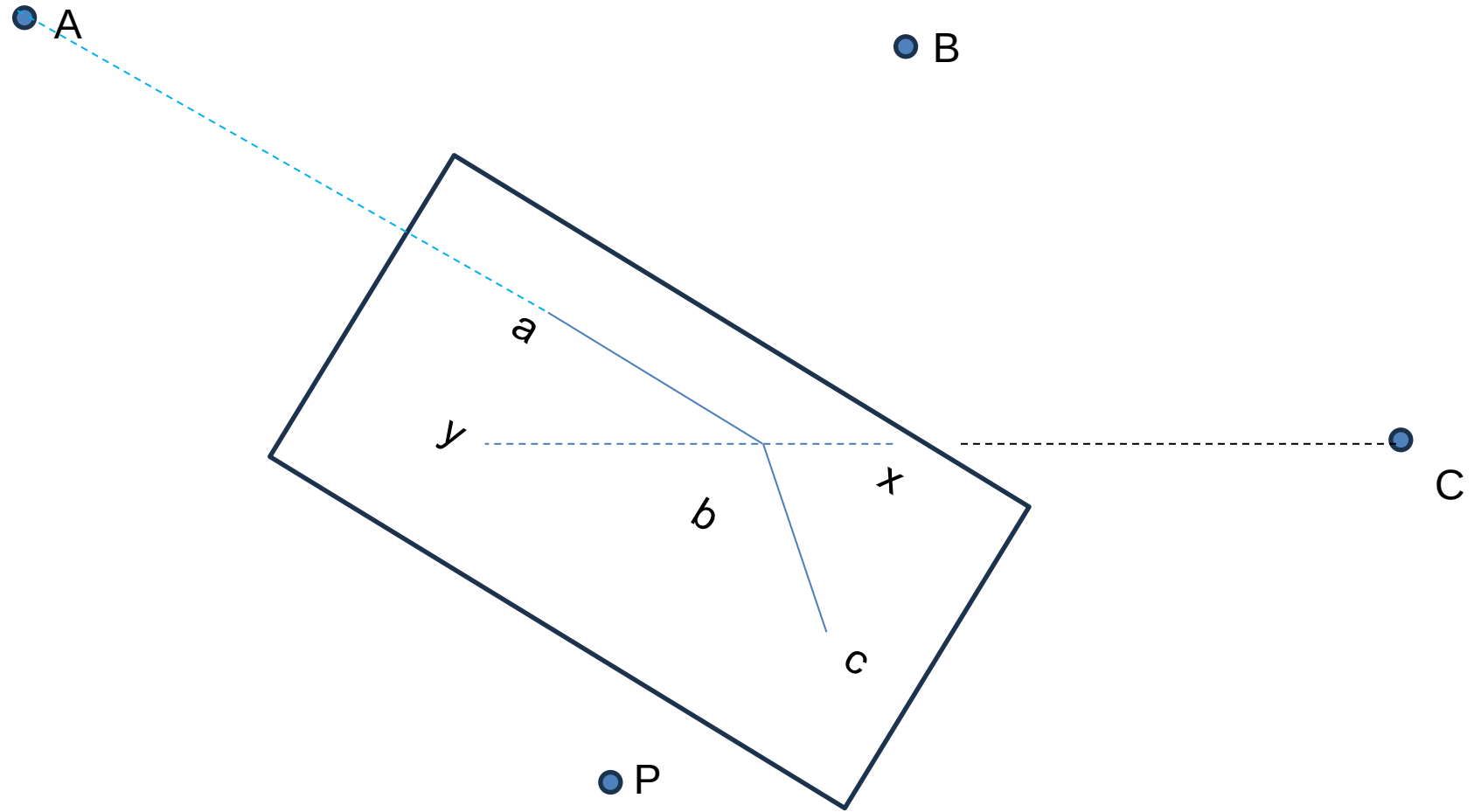
● A

● B



● C

● P

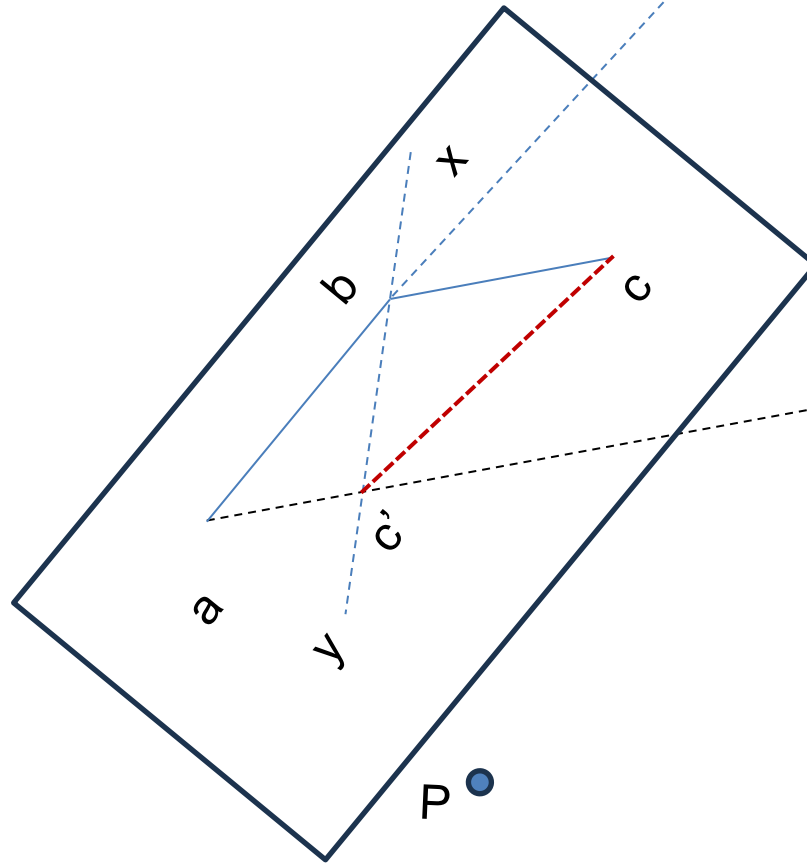


Step-1

• A

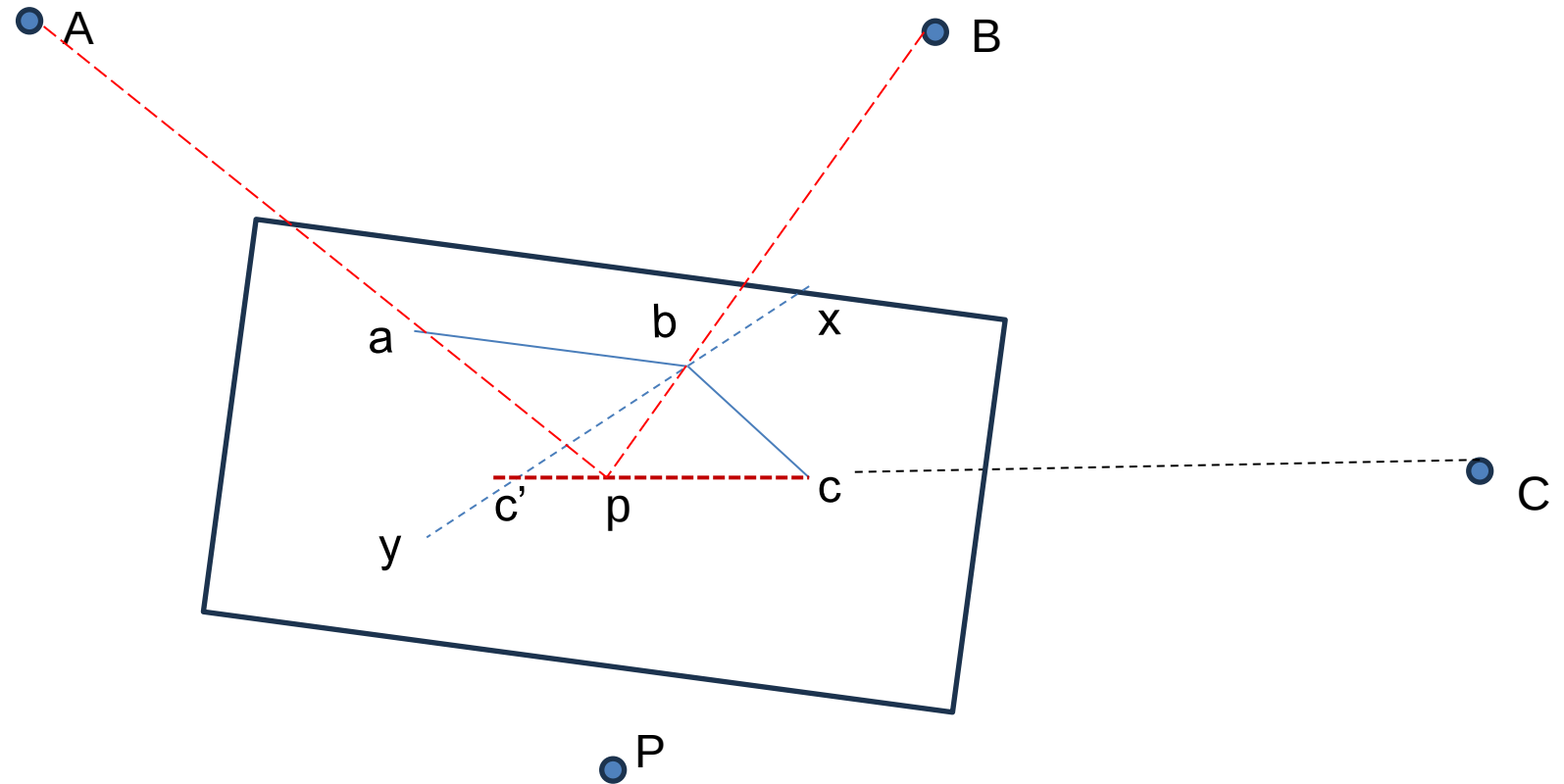
• B

C



P

Step-2



Step-3

